the five-minute film of the race all the way back to the two-minute mark). Similarly, since in the standard big bang theory gravity slows the expansion of space, from any point in the cosmic film we have to wind more than halfway back in time in order to halve the separation between two regions. And, as above, this means that even though the regions of space were closer together at earlier times, it was more difficult—not less—for them to influence each other and hence more puzzling—not less—that they somehow reached the same temperature.

Physicists define a region's *cosmic horizon* (or *horizon* for short) as the most distant surrounding regions of space that are close enough to the given region for the two to have exchanged light signals in the time since the bang. The analogy is to the most distant things we can see on earth's surface from any particular vantage point.[15] The horizon problem, then, is the puzzle, inherent in the observations, that regions whose horizons have always been separate—regions that could never have interacted, communicated, or exerted any kind of influence on each other—somehow have nearly identical temperatures.

The horizon problem does not imply that the standard big bang model is wrong, but it does cry out for explanation. Inflationary cosmology provides one.

In inflationary cosmology, there was a brief instant during which gravity was repulsive and this drove space to expand faster and faster. During this part of the cosmic film, you would have to wind the film *less* than halfway back in order to halve the distance between two regions. Think of a race in which Secretariat covers the first half of the course in two minutes and, because he's having the run of his life, speeds up and blazes through the second half in one minute. You'd only have to wind the three-minute film of the race back to the two-minute mark—less than halfway back—to see him at the course's halfway point. Similarly, the increasingly rapid separation of any two regions of space during inflationary expansion implies that halving their separation requires winding the cosmic film less—*much less*—than halfway back toward the beginning. As we go farther back in time, therefore, it becomes *easier* for any two regions of space to influence each other, because, proportionally speaking, there is more time for them to communicate. Calculations show that if the inflationaryexpansion phase drove space to